

Ryan Atkinson

GitHub: <https://github.com/Ryan-Atkinson87>

Portfolio: <https://portfolio.trivedev.uk>

LinkedIn: <https://www.linkedin.com/in/ryan-atkinson-804a5b1a8>

AI-First Software Engineer | Full-Stack & IoT Systems

Personal Statement

AI-first software engineer with hands-on experience delivering full-stack and IoT solutions using modern AI-assisted development workflows. I actively integrate AI tools into day-to-day engineering, using them to design, refine, and accelerate production-ready systems rather than as an afterthought.

I have built and deployed Raspberry Pi-based monitoring systems using Python and MQTT, integrating live telemetry into ThingsBoard, and developed containerised web applications using Django, React, Docker, and Linux-based infrastructure. My background in IT engineering and technical support has shaped a structured, systems-focused approach to troubleshooting, documentation, and continuous improvement.

I am motivated by ownership, clarity of thinking, and building maintainable systems that solve real operational problems.

Technical Skills

Languages: Python, JavaScript, C++, SQL

Frameworks & Libraries: Django, React (Vite), Bootstrap, MQTT, Node-RED

DevOps & Infrastructure: Docker, Linux CLI, systemd, Cloudflare Tunnel, Raspberry Pi

AI-Assisted Development: LLM-assisted coding workflows, structured prompt iteration, AI-supported debugging and refactoring

Databases: PostgreSQL, T-SQL

Other: REST APIs, MQTT protocols, basic OPC-UA integration

Technical Projects

Aquarium Monitoring Platform (2025 to present)

Designed and deployed a scalable IoT monitoring system using Python and MQTT to stream sensor telemetry from Raspberry Pi GPIO devices into ThingsBoard. Leveraged AI-assisted

development workflows to accelerate system design, debugging, and refactoring while maintaining production-level reliability.

Implemented secure remote access via Cloudflare Tunnel and structured the project with modular components and documentation to support future productisation and reuse.

GitHub: https://github.com/Ryan-Atkinson87/trive_aquasense

Full-Stack Breeder Network Platform (2024 to present)

Built a Django backend and React (Vite) frontend to connect fish breeders and local stores. Hosted in a Docker container on Raspberry Pi with SSL. Designed API endpoints, user workflows, and database models with future extensibility in mind.

GitHub: <https://github.com/Ryan-Atkinson87/TriveDevHome>

Raspberry Pi Robot (2024 to present)

Constructed a programmable robot with custom 3D-printed components. Programmed in Python with PyGame handling user control. Plans include camera feed, controller input, and arm articulation.

Portfolio Website

Developed a clean, accessible portfolio using React with a contact form using the Resend API.

Live: <https://portfolio.trivedev.uk/> | GitHub: <https://github.com/Ryan-Atkinson87/MyPortfolio>

AI-Integrated Development Approach

In recent projects, AI-assisted tools are embedded into my workflow for architecture exploration, structured refactoring, test scaffolding, and debugging. I treat AI as a collaborative engineering tool and actively refine prompts and context to improve output quality and maintain control over system design.

Courses and Certifications

Codecademy

- Learn C++
- Learn OOP with Python
- Learn Git and GitHub
- Full-Stack Engineer Path (in progress)
- Exploring C# and .NET Framework

Education

BSc Computing and IT (Software), 2022 to present

Open University

Modules completed: TM111, TM112, Discovering Mathematics, Technologies in Practice, Object-Oriented Java Programming.

Practical exposure to networking, Linux OS, and Python-based robotics.

BSc Marine Biology, 2007 to 2010

Bangor University

Covered marine ecology, biodiversity, conservation, fieldwork, and biostatistics.

Professional Experience

Operations Technician, CDL Group (Sep 2025 – Present)

- Investigate and resolve incidents relating to aggregator XML file ingestion and processing within a live production environment.
- Diagnose portal login and registration failures involving AWS-hosted authentication workflows and backend service integrations.
- Resolve back-office platform failures across both modern web applications and legacy software systems.
- Perform configuration updates, log analysis, and structured diagnostics to identify root causes of integration and data-processing errors.
- Execute targeted SQL queries to trace data discrepancies, validate transactional states, and support incident resolution.
- Work cross-functionally to communicate technical findings clearly and ensure stable restoration of services.

IT Engineer (Graduate), RLC Callender (Nov 2024 to Sep 2025)

- Maintained and enhanced internal VB.NET web applications, contributing production updates to a live business-critical system
- Used AI-assisted workflows to build small internal tools and improve day-to-day development efficiency
- Deployed and configured 11 Raspberry Pi units for machine data monitoring using Python and MQTT, integrating telemetry into central dashboards
- Contributed to Appian low-code rollout, gaining exposure to enterprise workflow design and solution architecture principles
- Supported internal databases and LMS platforms with a structured, process-driven engineering approach

Service Desk Analyst, Giacom Ltd (2023 to 2024)

- Used SQL to validate and correct billing issues
- Managed Azure user accounts and onboarding workflows
- Gained exposure to backend and API support tickets

Service Desk Analyst, Daisy Corporate Services (2022 to 2023)

- Troubleshoot broadband and connectivity issues and Cisco routers using SSH and Telnet
- Worked with ISPs and escalated faults efficiently

Sheet Metal Engineer, 2012 to 2022

- Roles included CAD Engineer, Estimator, and Supervisor
- Worked with SolidWorks, laser programming, and technical customer liaison

- Built strong analytical skills, team leadership, and hands-on problem solving

Interests

Web Development – Enjoy designing clean, user-first interfaces

Tech Trends – Follow cloud, automation, and open-source ecosystems

Fish Keeping – Maintain aquariums which inspire real-world project ideas

Triathlon – Committed to consistency, goal setting, and personal challenge

References

Available on request